

Respected Reviewers,

We Authors of the submission 237, Firstly thankful for your reviews to our paper. We tried to address the comments and took actions against them.

1. **Methodology Enhancement:** It might be beneficial to employ computer vision techniques to enhance images for object classification, and exploring deep learning methods could improve accuracy, even with small datasets.

We appreciate the reviewer's suggestion to explore computer vision techniques for enhancing object classification. While template matching using Chamfer distance offers flexibility in handling object changes without relabeling, we acknowledge the potential benefits of deep learning methods for improved accuracy, especially with small datasets. To address this, we plan to investigate hybrid approaches that combine template matching with deep learning techniques, such as using template matching for initial detection and deep learning for more accurate classification. Additionally, in our future work, we will explore data augmentation strategies to increase the diversity of our training data and potentially leverage transfer learning to improve model performance.

2. **Template and Reference Consistency:** Please consider reviewing the template for equation referencing. Using '(no.)' instead of 'Eq (no.)' and starting sentences with 'Equation (no.)' may enhance clarity. In addition, it might be helpful to ensure reference formatting is consistent, including the use of 'pp' for page numbers and the correct format for conference papers.

We appreciate the reviewer's suggestion to consider the change of equation formatting and clear references including pp(page numbers). Equation formatting has changed in the document and action taken on reference to include the page numbers. And marked in red color as suggested by review form guidelines.

3. **Reference Update:** The author(s) might want to reevaluate references [1] and [2], as they could be outdated.

We appreciate the reviewer's suggestion to reevaluate references [1] and [2]. We considered this comment and rearranged our related work section from [3] reference.

4. **Clarify Experimental Setup:** Provide a clearer experimental setup, specifying the number of sizes and bolt positioning scenarios in each bin. In addition, include R&R gauge measurements to assess the consistency of your results.

We appreciate the reviewer's suggestion, we have explained about experimental setup under the section experimental setup and methodology. Added R&R gauge measurements table for 4 trials under the section Results and discussions.

5. **Conduct Error Analysis: Perform error analysis under varying lighting conditions to identify the most effective setup.**

We appreciate the reviewer's suggestion, to improve the system's robustness to real-world lighting variations, we initially investigated the impact of different lighting conditions in our experimental setup. However, our experiments with varying lighting intensities, color temperatures did not reveal significant differences in the system's performance. Based on these findings, we concluded that our core algorithms and techniques were sufficiently robust to handle variations in lighting. As a result, we removed information regarding varying lighting conditions from our manuscript to streamline the presentation and focus on other key aspects of the system. While we have demonstrated the system's resilience to lighting changes in our experiments, we acknowledge that real-world scenarios can present unique challenges. Therefore, we plan to continue monitoring the system's performance in deployed environments to identify any potential limitations and make necessary adjustments in our future work.

6. **Position Estimation Clarity: Clarify how the estimated position is computed.**

We have considered the reviewer's suggestion, including equations (3), (4), (5) which will give clarification on how we calculated the position of the bolt.

7. **Template Matching Considerations: Be aware of the limitations of template matching with Chamfer distance. While it can classify specific sizes, it may not suit real-world applications with diverse sizes and types. Consider more flexible approaches for better accuracy.**

We acknowledge the limitations of template matching with Chamfer distance, especially regarding size sensitivity. To address these concerns, we plan to incorporate image normalization techniques, explore alternative distance metrics like Hausdorff distance, and investigate feature-based matching approaches. We believe these strategies will enhance the robustness and accuracy of our method for real-world applications. By carefully considering these strategies and incorporating them into our future work, we can improve the effectiveness of our template matching approach and address the reviewer's concerns.

8. **Summarize Related Works: Summarize the related works section in a concise table for Clarity.**

While we appreciate the suggestion to summarize the related works in a concise table, we believe that a narrative approach provides a more comprehensive and nuanced understanding of the existing literature. A table, while helpful for quick reference, might overlook the historical context, interconnections between works, and the subtle nuances that distinguish our work from previous research. By presenting the related works in a narrative format, we can effectively convey the evolution of the field and highlight the unique contributions of our research. We believe this approach offers a more holistic understanding of the relevant literature and strengthens the overall argument of our paper.

9. **Index Terms Specification: Ensure index terms consist of 3-5 words for optimal clarity.**

We have included only 4 essential index terms related to the present work. And marked in red color as suggested by review form guidelines.

10. **It is suggested to cite the following references to enhance the background of your paper.**

Phan Gia Luan and Nguyen Truong Thinh, "Overconstrained Cable-Driven Parallel Manipulator's Statics Analysis Based on Simplified Static Cable Model," *International Journal of Mechanical Engineering and Robotics Research*, Vol. 11, No. 3, pp. 159-165, March 2022. DOI: 10.18178/ijmerr.11.3.159-165.

Nguyen Van Quyen, Bui Thi Hai Linh, Duong Quoc Tuan, Nguyen Trung Thanh, and Nguyen Hong Quang, "Dynamic Stability Control and Calculating Inverse Dynamics of a Single-link Flexible Manipulator," *International Journal of Mechanical Engineering and Robotics Research*, Vol. 12, No. 4, pp. 200-209, July 2023. DOI: 10.18178/ijmerr.12.4.200-209.

Mohammed J. Mohammed, Basma A. Abbas, and Sunddus A. Al-Azawiey, "Flexible Underwater Manipulator Modeling Using Intelligent Method" *International Journal of Mechanical Engineering and Robotics Research*, Vol. 8, No. 6, pp. 860-866, November 2019. DOI: 10.18178/ijmerr.8.6.860-866.

We appreciate the reviewer's suggestion to consider the references. However, after careful examination, we believe that these references are not directly relevant to our research. Our primary focus is on automation of industrial assembly with manipulators using vision-based methods. The suggested references primarily address the area of designing and static and dynamic analysis of manipulators. While these areas are undoubtedly important, they do not directly contribute to the specific problems or questions we are investigating in our study. Therefore, we believe that including these references would not significantly enhance the background or context of our research.